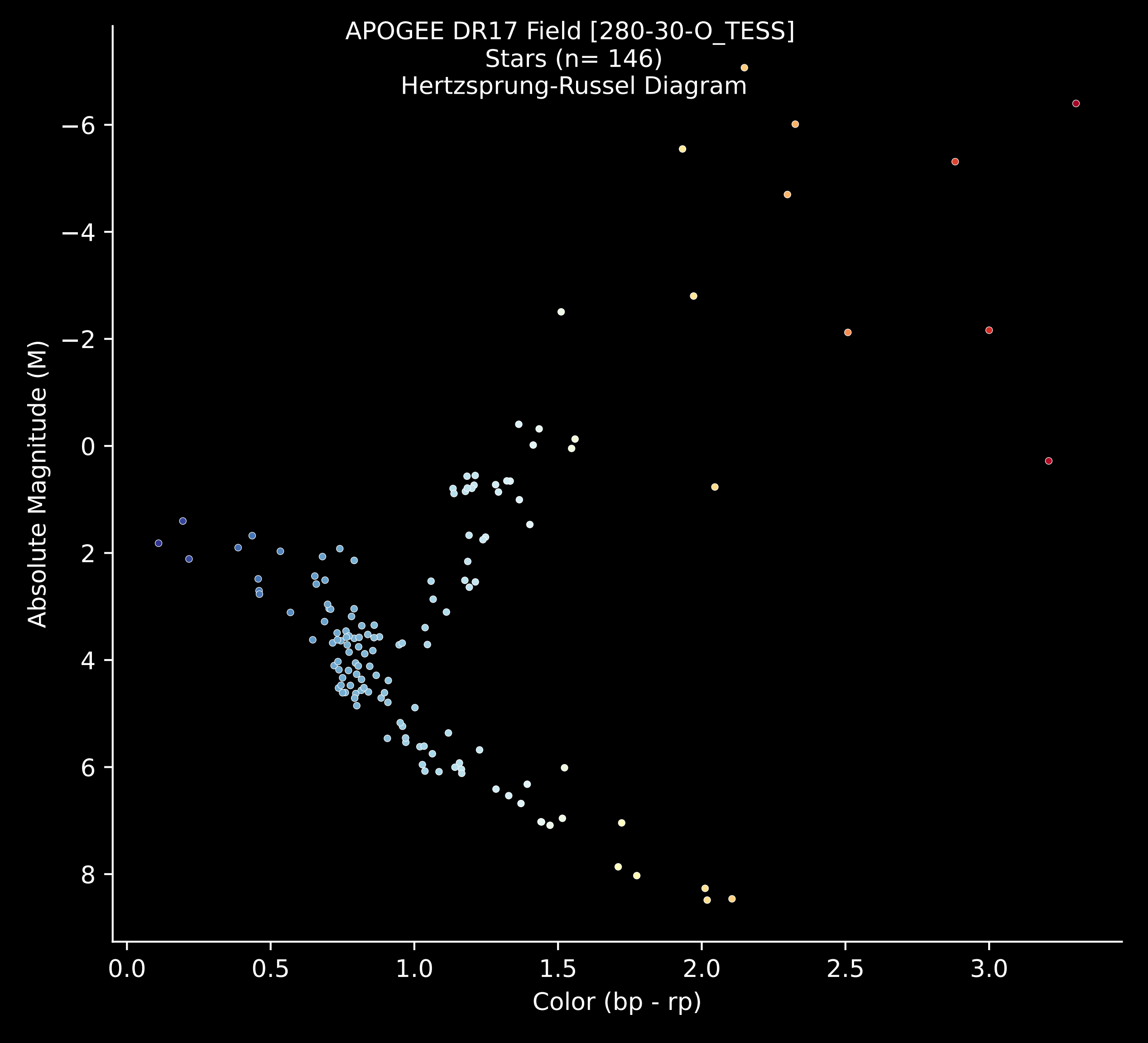
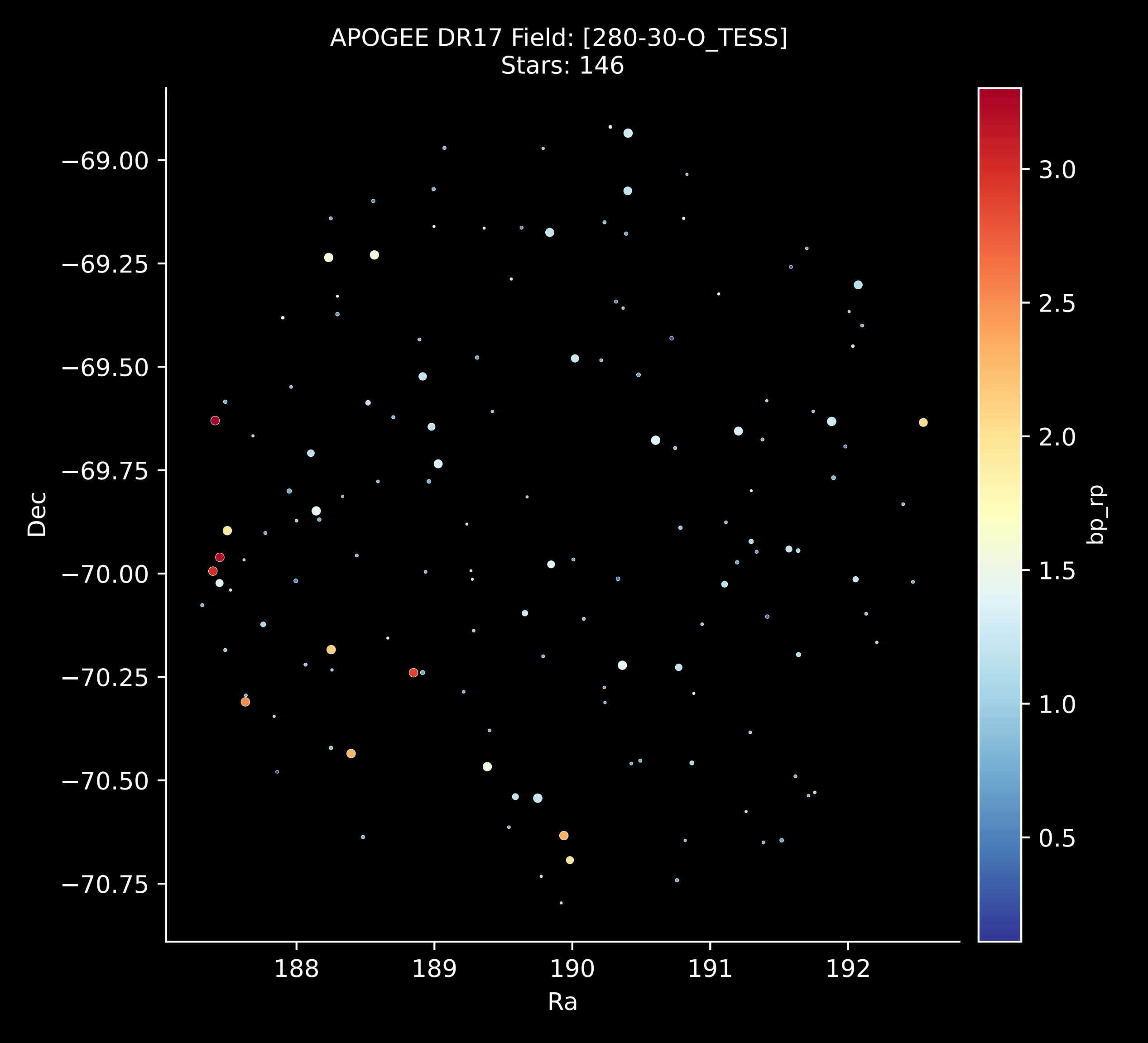
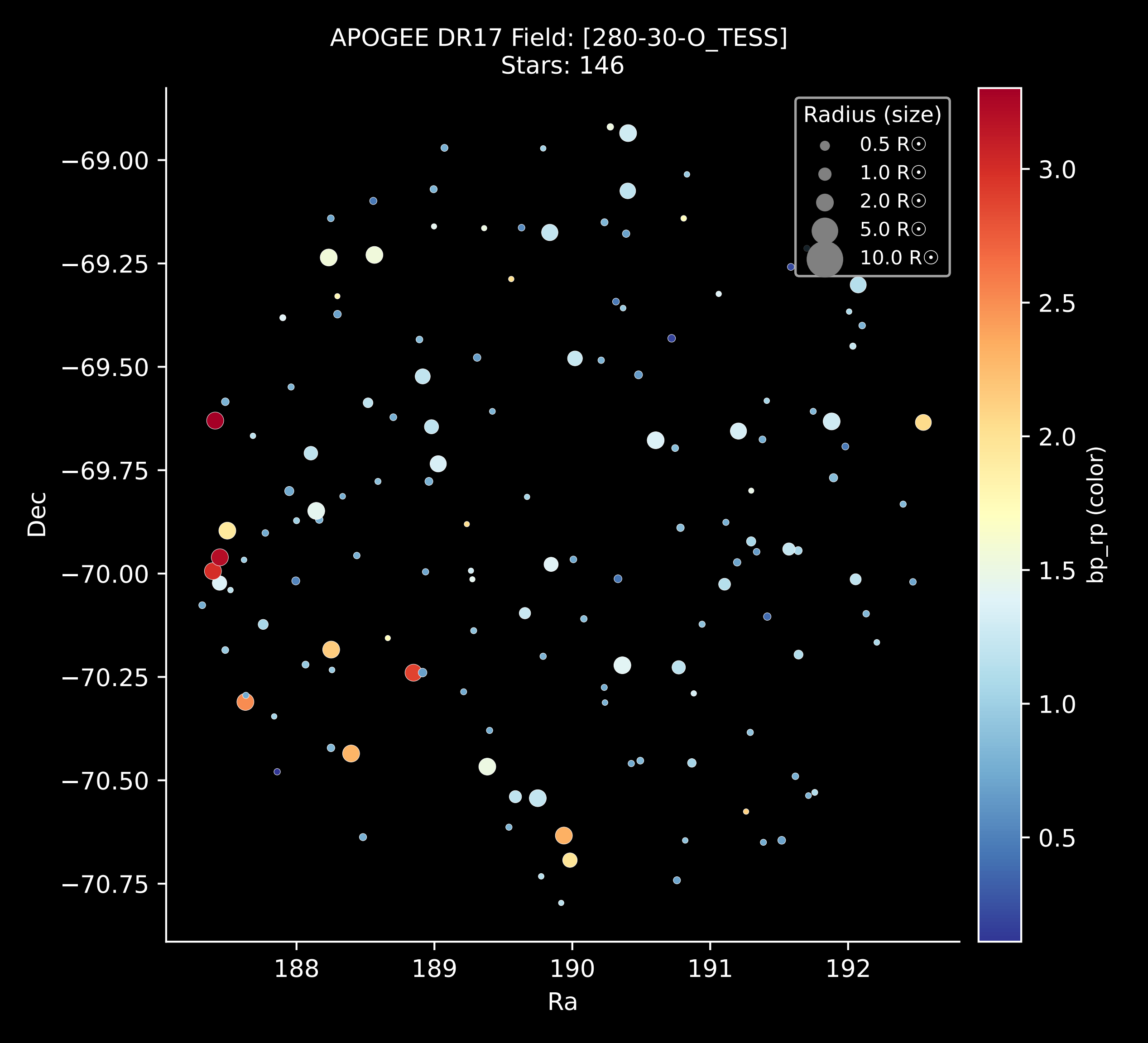
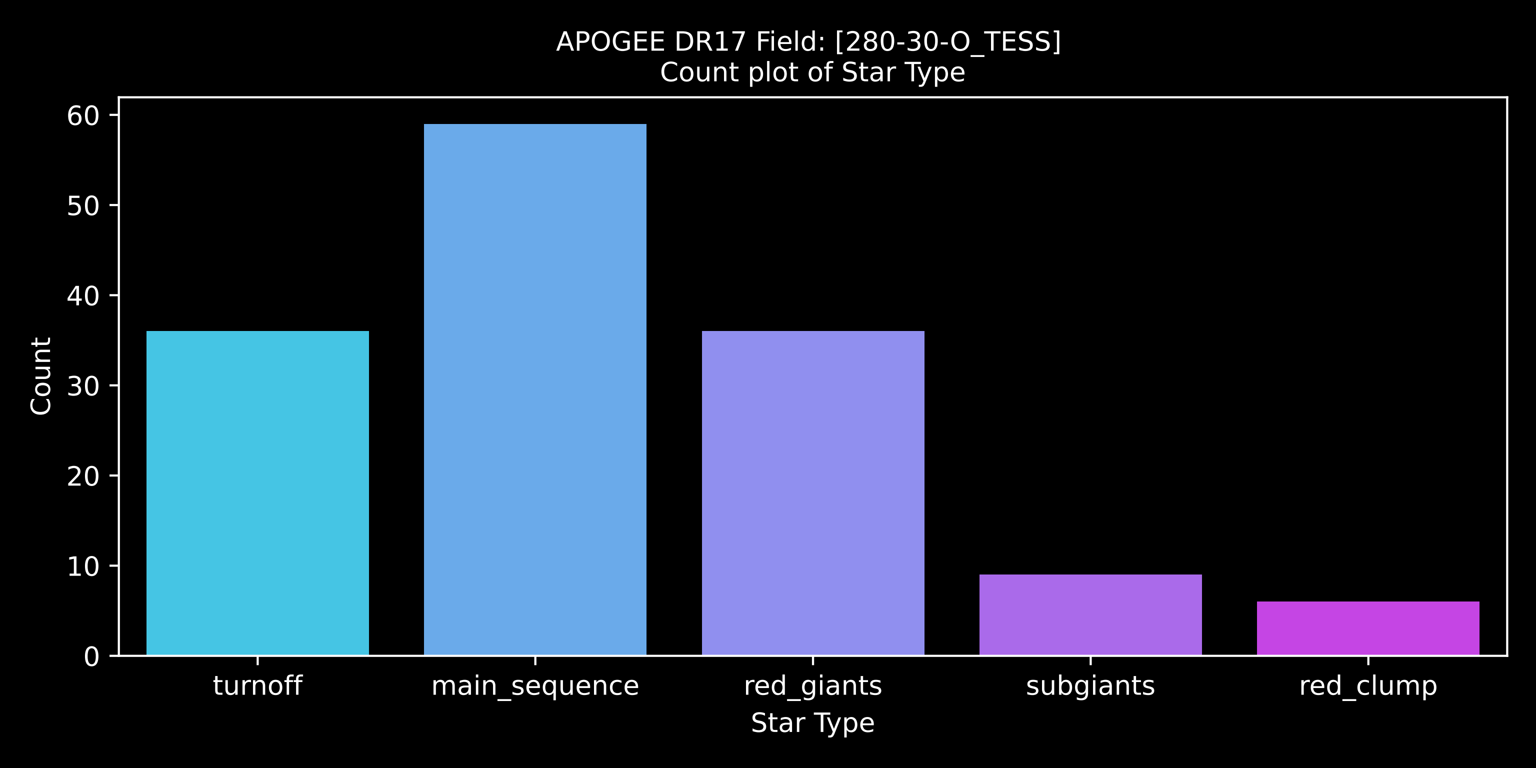
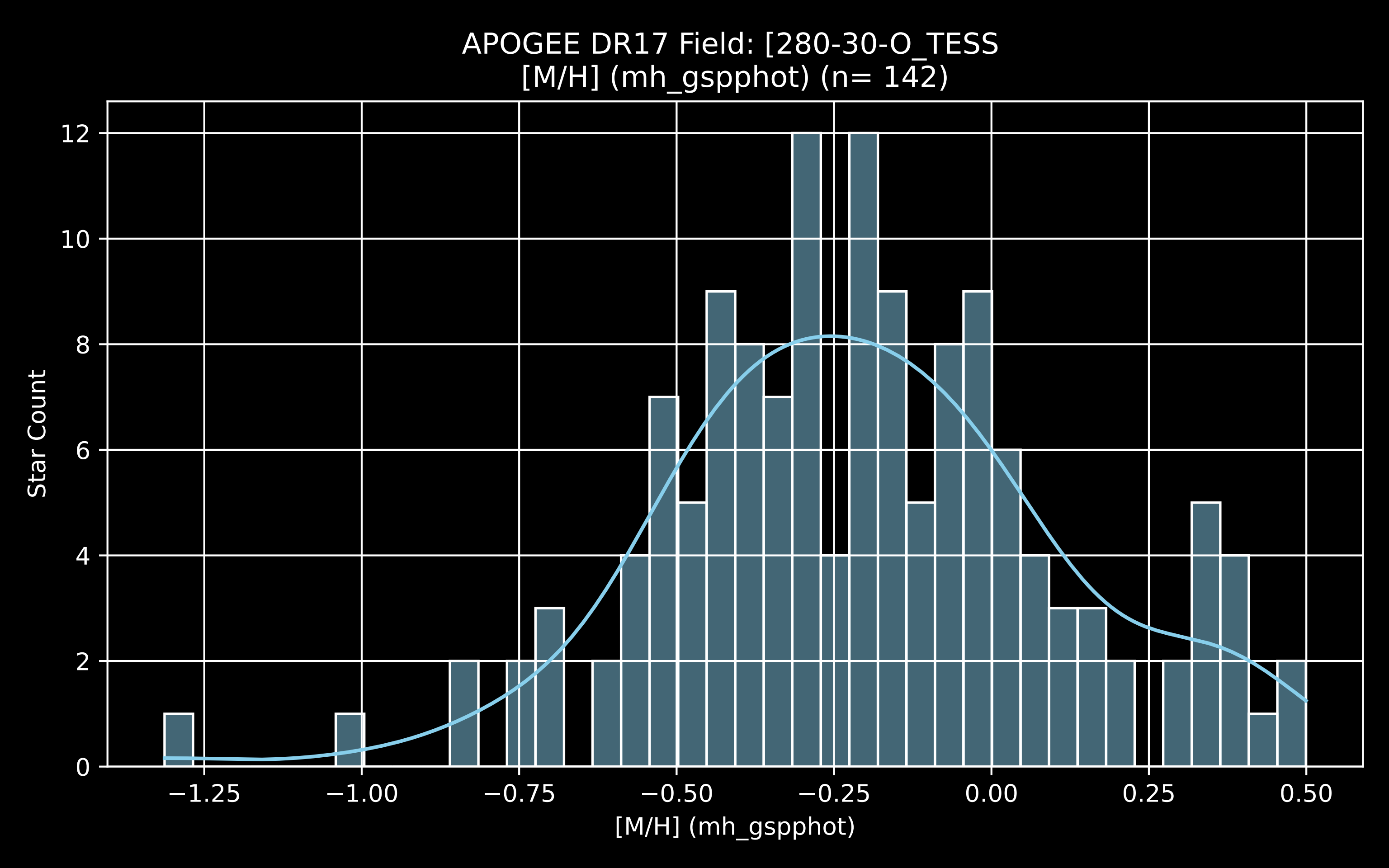

APOGEE DR17 Field [280-30-O_TESS]
Stars (n= 146)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−6
−4
−2
0
2
4
6
8
0.0
0.5
1.0
1.5
2.0
2.5
3.0


APOGEE DR17 Field: [280-30-O_TESS]
Stars: 146
Dec
Ra
bp_rp
−69.00
−69.25
−69.50
−69.75
−70.00
−70.25
−70.50
−70.75
188
189
190
191
192
3.0
2.5
2.0
1.5
1.0
0.5

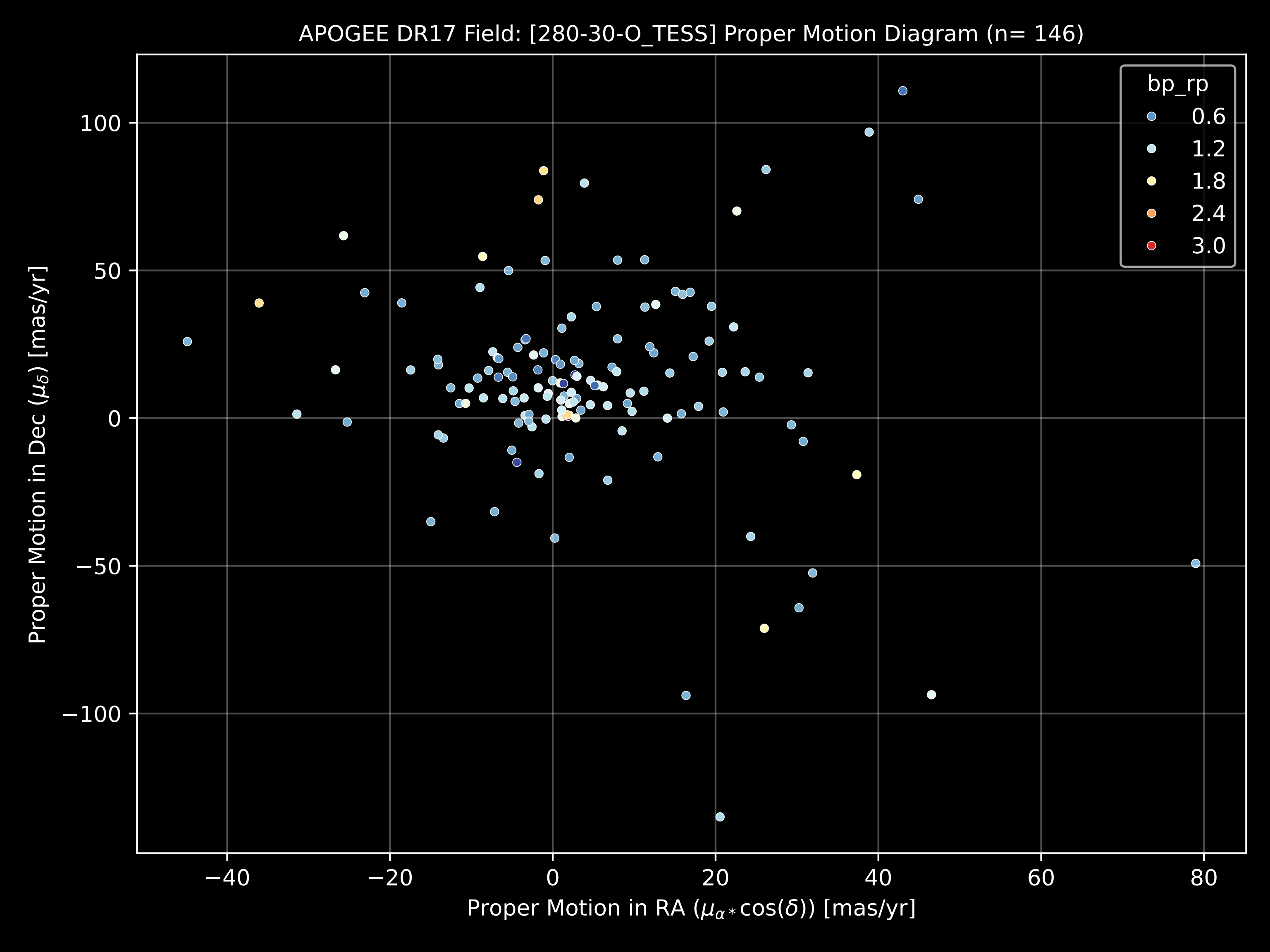


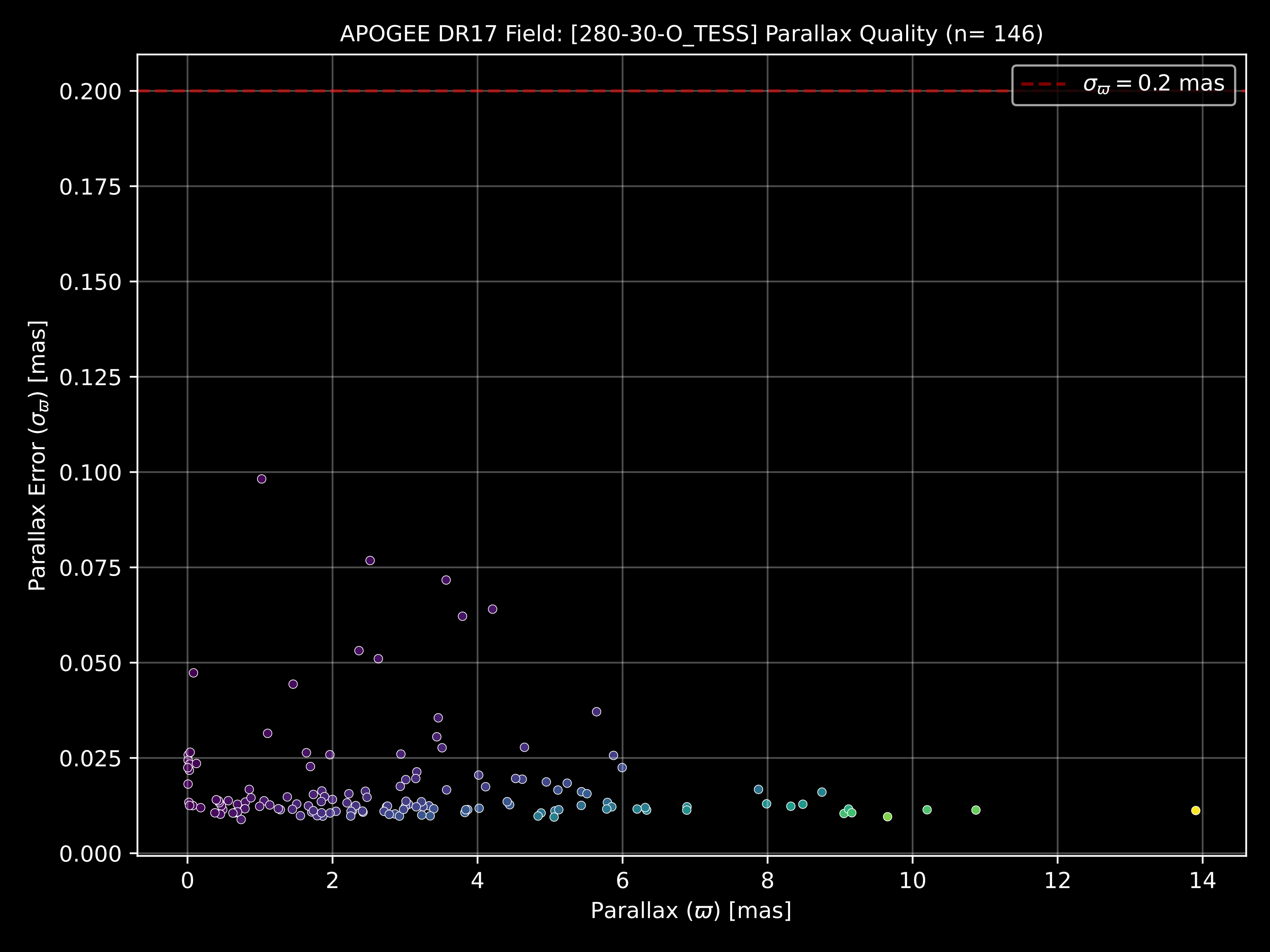
APOGEE DR17 Field: [280-30-O_TESS]
Stars: 146
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
−69.00
−69.25
−69.50
−69.75
−70.00
−70.25
−70.50
−70.75
Dec
188
189
190
191
192
Ra
3.0
2.5
2.0
1.5
1.0
0.5
bp_rp (color)


APOGEE DR17 Field: [280-30-O_TESS]
Count plot of Star Type
Count
60
50
40
30
20
10
0
turnoff
main_sequence
red_giants
subgiants
red_clump
Star Type

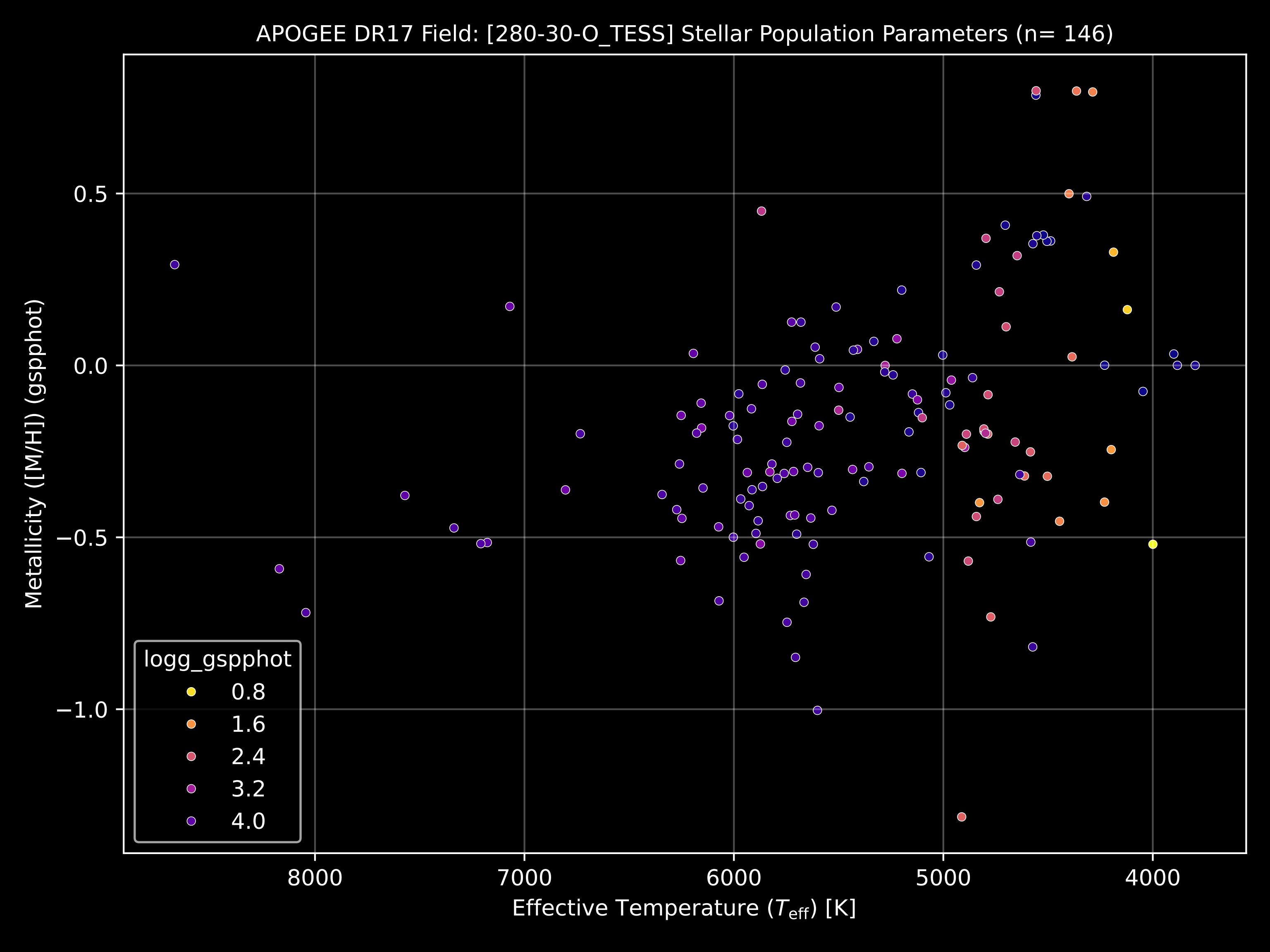


APOGEE DR17 Field: [280-30-O_TESS
[M/H] (mh_gspphot) (n= 142)
Star Count
[M/H] (mh_gspphot)
0
2
4
6
8
10
12
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50

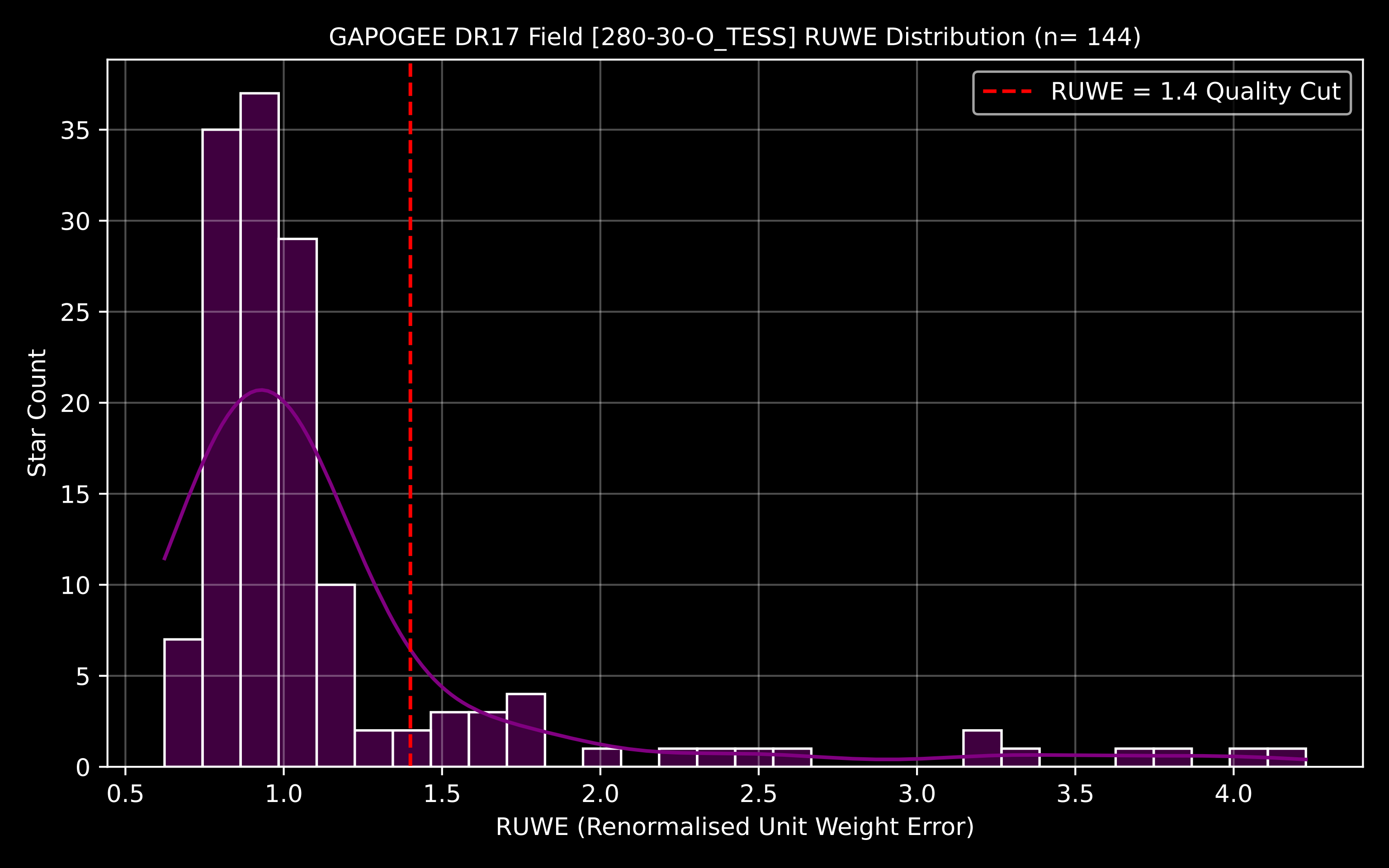

APOGEE DR17 Field: [280-30-O_TESS] Proper Motion Diagram (n= 146)
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
bp_rp
0.6
1.2
1.8
2.4
3.0
100
50
0
−50
−100
−40
−20
0
20
40
60
80



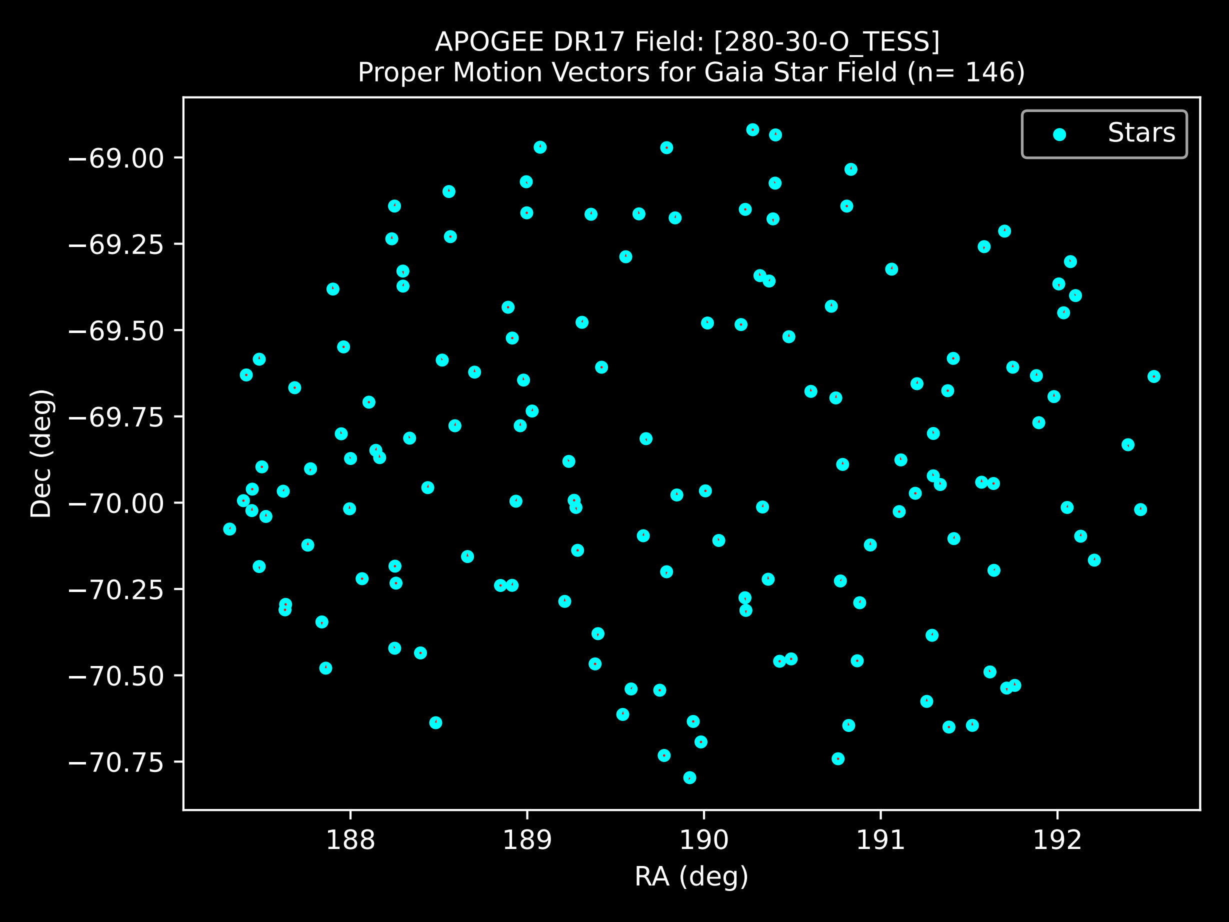
APOGEE DR17 Field: [280-30-O_TESS] Parallax Quality (n= 146)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0
2
4
6
8
10
12
14

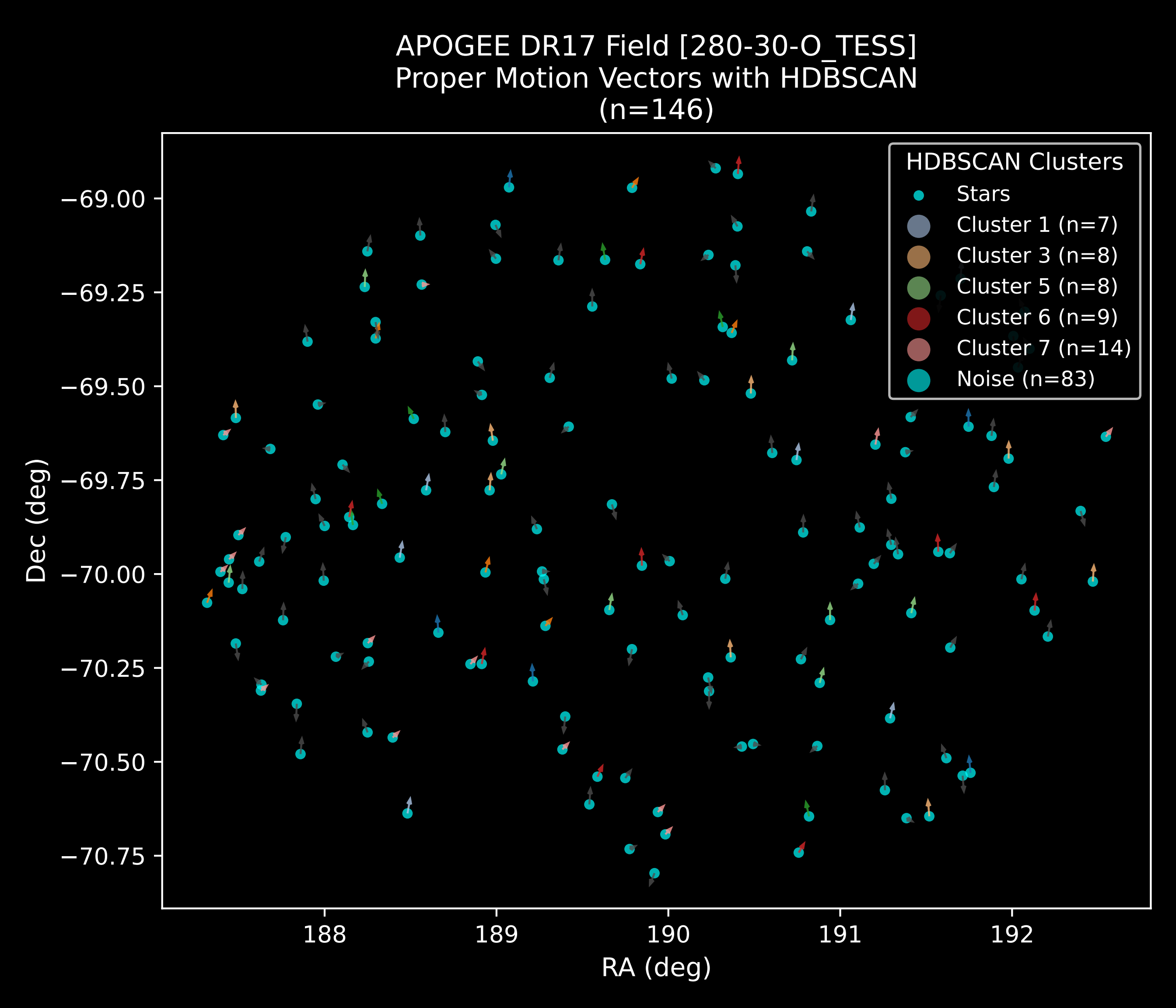


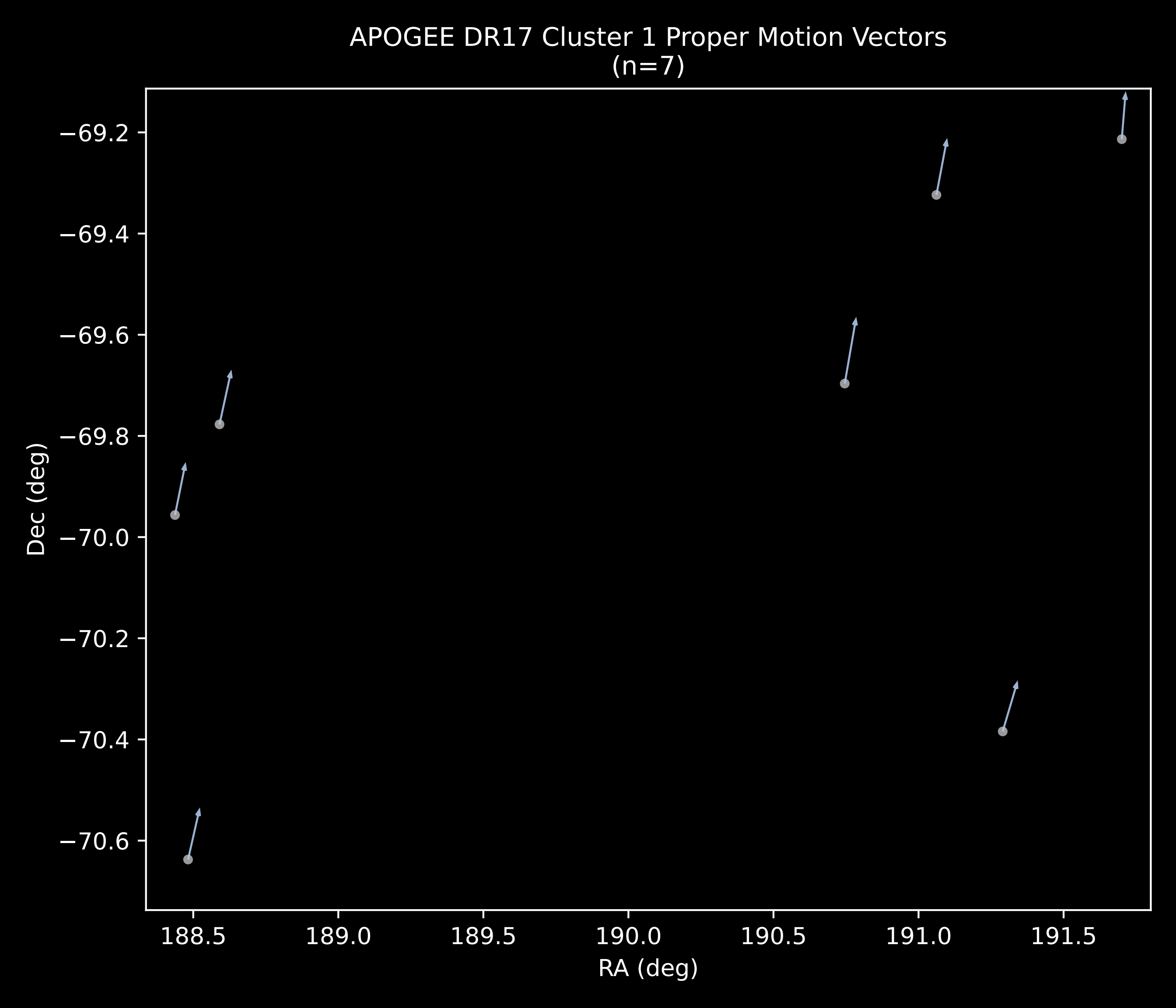
APOGEE DR17 Field: [280-30-O_TESS] Stellar Population Parameters (n= 146)
Metallicity ([M/H]) (gspphot)
Effective Temperature (T_{eff}) [K]
0.5
0.0
−0.5
−1.0
8000
7000
6000
5000
4000
logg_gspphot
0.8
1.6
2.4
3.2
4.0

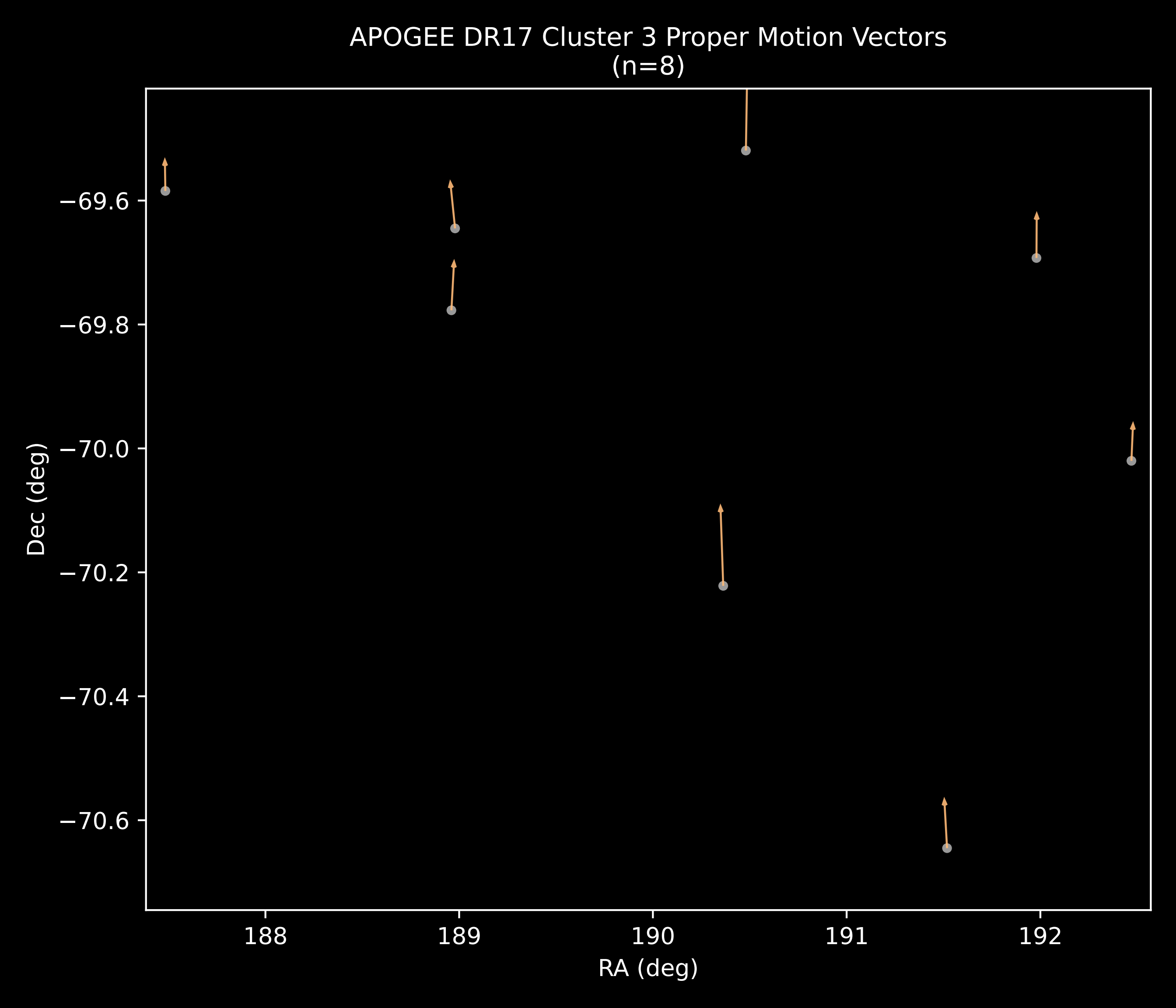


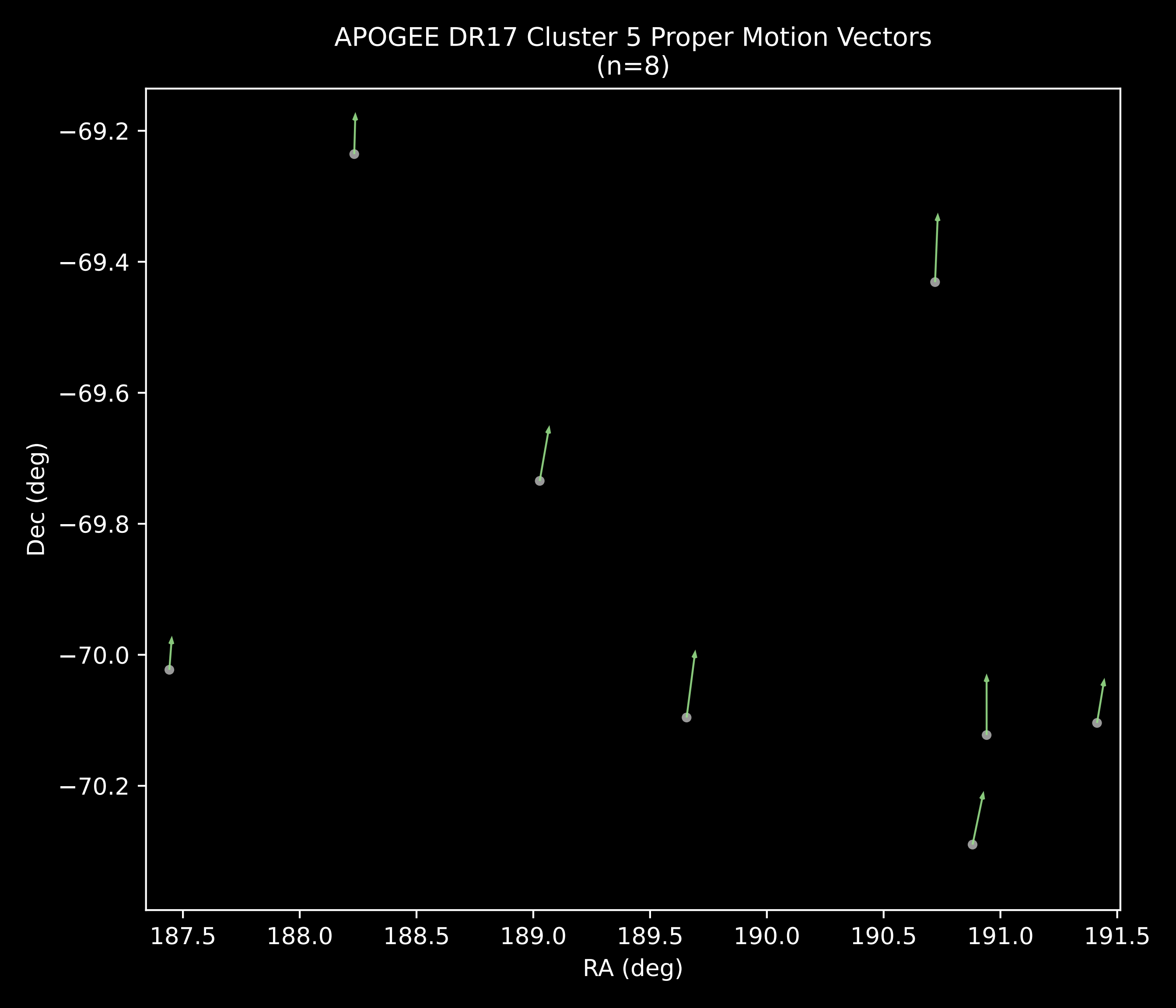
GAPOGEE DR17 Field [280-30-O_TESS] RUWE Distribution (n= 144)
RUWE = 1.4 Quality Cut
Star Count
RUWE (Renormalised Unit Weight Error)
0
5
10
15
20
25
30
35
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

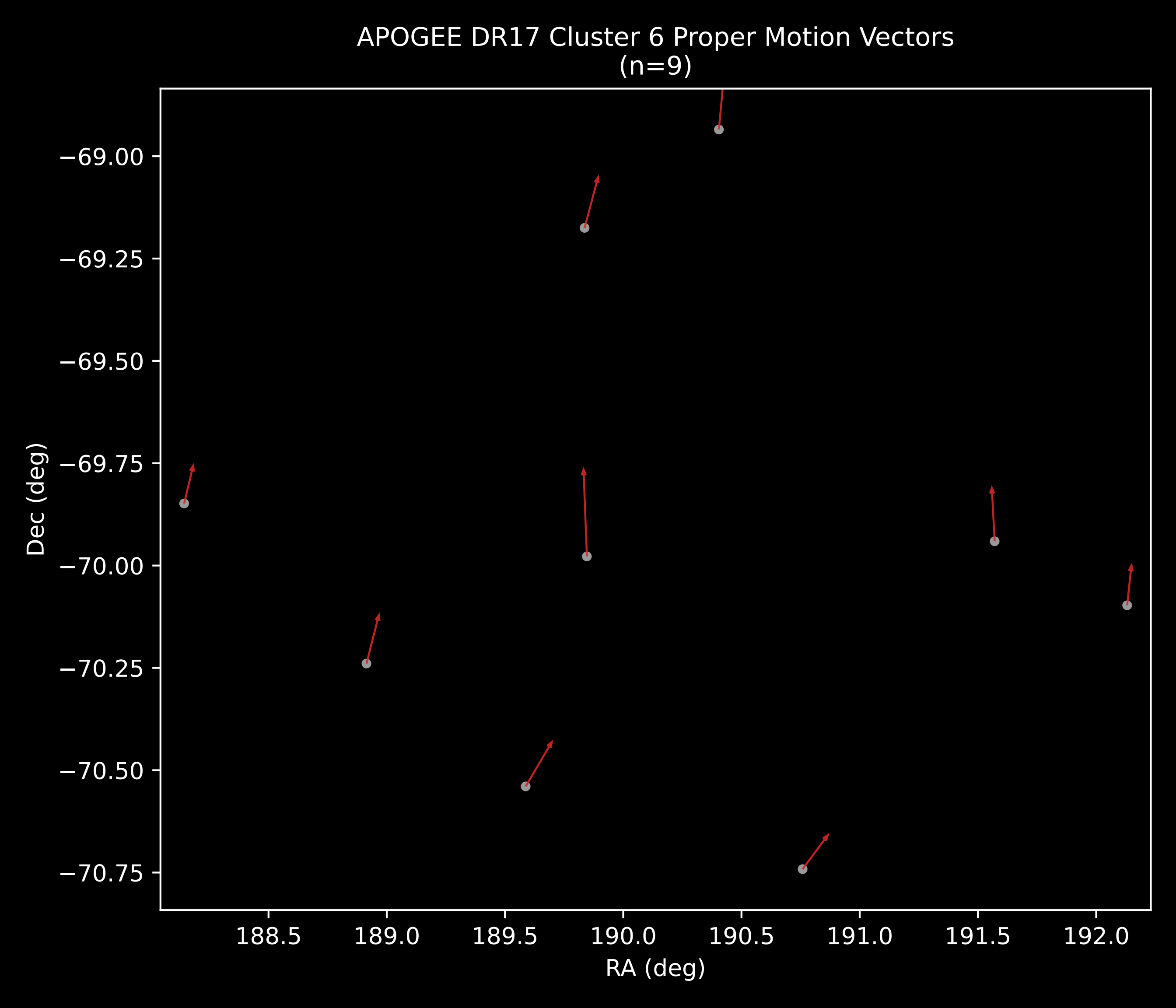

APOGEE DR17 Field: [280-30-O_TESS]
Proper Motion Vectors for Gaia Star Field (n= 146)
Stars
−69.00
−69.25
−69.50
−69.75
−70.00
−70.25
−70.50
−70.75
Dec (deg)
188
189
190
191
192
RA (deg)

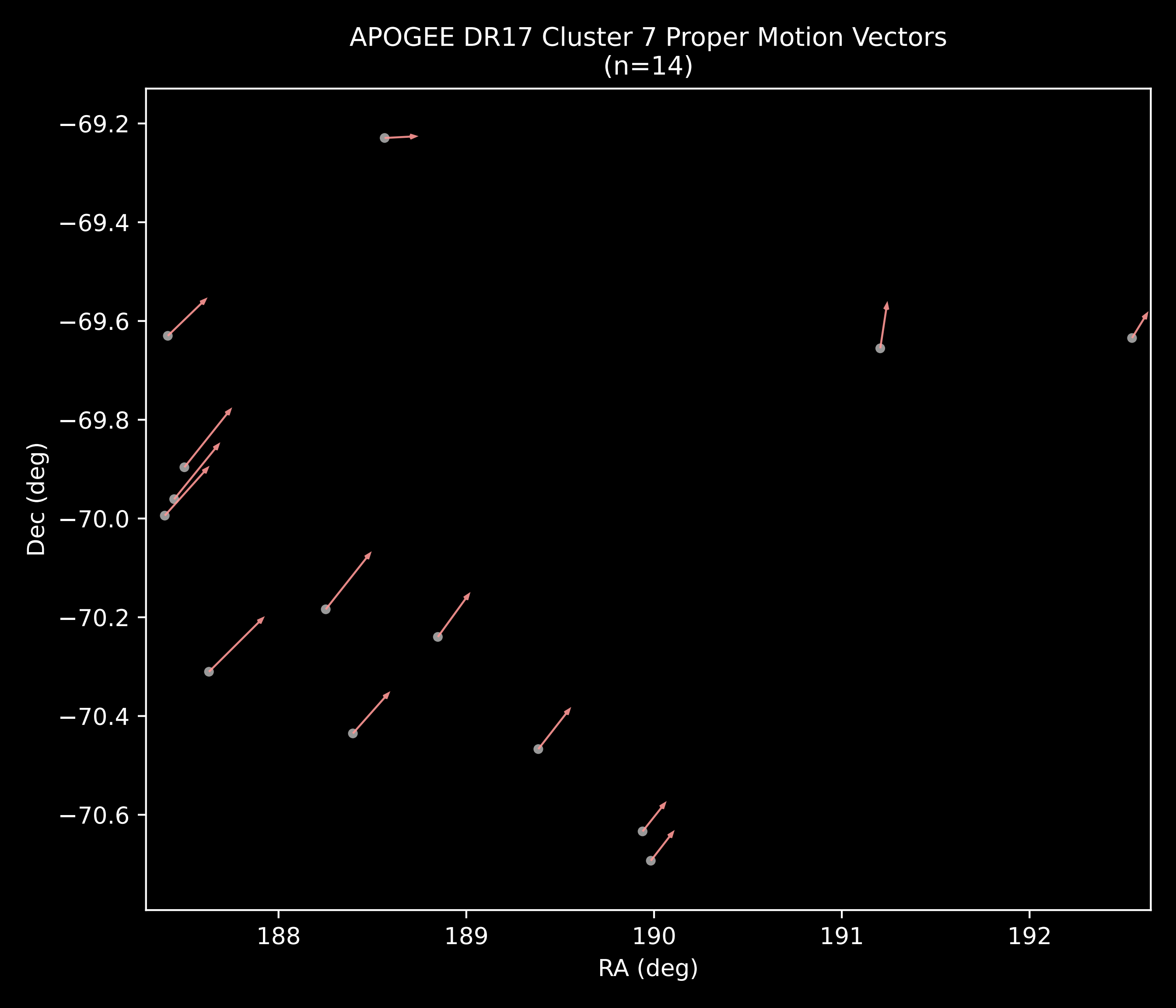

APOGEE DR17 Field [280-30-O_TESS]
Proper Motion Vectors with HDBSCAN
(n=146)
HDBSCAN Clusters
Stars
Cluster 1 (n=7)
Cluster 3 (n=8)
Cluster 5 (n=8)
Cluster 6 (n=9)
Cluster 7 (n=14)
Noise (n=83)
Dec (deg)
−69.00
−69.25
−69.50
−69.75
−70.00
−70.25
−70.50
−70.75
RA (deg)
188
189
190
191
192

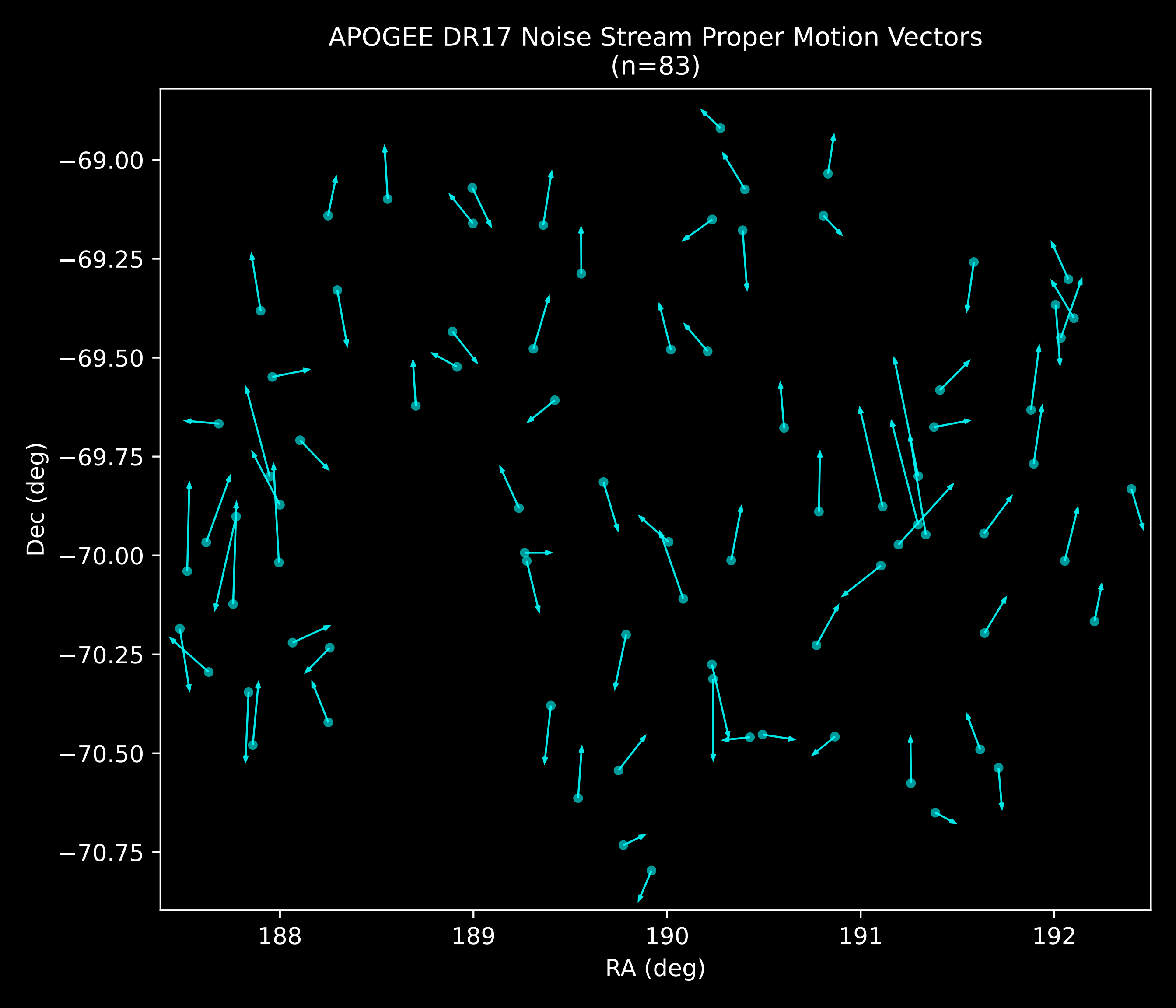

APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=7)
Dec (deg)
RA (deg)
−69.2
−69.4
−69.6
−69.8
−70.0
−70.2
−70.4
−70.6
188.5
189.0
189.5
190.0
190.5
191.0
191.5


APOGEE DR17 Cluster 3 Proper Motion Vectors
(n=8)
Dec (deg)
−69.6
−69.8
−70.0
−70.2
−70.4
−70.6
188
189
190
191
192
RA (deg)


APOGEE DR17 Cluster 5 Proper Motion Vectors
(n=8)
−69.2
−69.4
−69.6
−69.8
−70.0
−70.2
Dec (deg)
187.5
188.0
188.5
189.0
189.5
190.0
190.5
191.0
191.5
RA (deg)


APOGEE DR17 Cluster 6 Proper Motion Vectors
(n=9)
Dec (deg)
−69.00
−69.25
−69.50
−69.75
−70.00
−70.25
−70.50
−70.75
RA (deg)
188.5
189.0
189.5
190.0
190.5
191.0
191.5
192.0


APOGEE DR17 Cluster 7 Proper Motion Vectors
(n=14)
−69.2
−69.4
−69.6
−69.8
−70.0
−70.2
−70.4
−70.6
Dec (deg)
188
189
190
191
192
RA (deg)


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=83)
Dec (deg)
RA (deg)
−69.00
−69.25
−69.50
−69.75
−70.00
−70.25
−70.50
−70.75
188
189
190
191
192